

Synthetic Data Evaluation Report

Sample 3 — Synthetic vs Real (metrics supplied)

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Executive Summary — Quick Read (1 minute)

Overall Quality Score (0–100): 94.55

This synthetic sample shows **strong overall utility** and excellent privacy indicators on the checks performed. Distributional fidelity and summary statistics are highly preserved, while privacy tests (duplicate, copy-risk, and re-identification) show no direct matches. A strong classifier signal was observed (high detectability AUC) — this indicates the synthesizer preserved structure that a model can exploit to distinguish records. That signal is actionable and can be reduced quickly if the client requires stricter indistinguishability.

Top highlights

- **Distribution match:** KS/JS/Wasserstein friendly scores are high (KS score **93.11**, JS score **98.47**, Wasserstein score **99.71**), indicating feature shapes and supports are well aligned.
- **Moments preserved:** Mean preservation = **98.65** and Std preservation = **98.91** — excellent preservation of central tendency and spread.
- **Missingness preserved:** Missingness match = **100%**.
- **Privacy posture:** Exact-duplicate fraction = **0%**; copy-risk fraction = **0%**; re-identification match fraction = **0%** on the tested quasi-identifiers.
- **High uniqueness and coverage:** Synthetic uniqueness **99.4%**; real-range coverage **95.08%**.
- **Correlation preserved:** Correlation preservation score **99.29** and Spearman preservation **98.10**.

Client-facing Summary of Results

Below are the measured values and short, plain-English interpretations. Numbers are shown exactly as supplied.

Metric	Value	One-line interpretation
Feature distribution (KS mean)	0.068875	KS-based metric shows good distributional match (friendly score 93.11/100).
Feature JS divergence (mean)	0.010592	JS divergence is low — shapes align well (98.47/100).
Wasserstein distance (mean)	0.016173	Very small earth-mover distance — support and mass align (99.71/100).
Mean preservation (relative avg error)	0.013499 (rel)	Mean preservation score: 98.65/100 — averages preserved.
Std (spread) preservation (relative avg error)	0.010868 (rel)	Std preservation score: 98.91/100 — spread preserved.
Median absolute difference (avg)	0.003225 (raw)	Medians are nearly identical.
10th percentile absolute diff (avg)	0.000257 (raw)	Very small lower-tail deviation.
90th percentile absolute diff (avg)	0.025437 (raw)	Small upper-tail deviation.
Missingness pattern difference (mean)	0.0	Missingness match: 100% (perfect).
Categorical JS mean	N/A	No categorical columns in this input (N/A).
Top category match ratio	N/A	N/A.
Category coverage mean	N/A	N/A.
Correlation matrix Frobenius difference (raw)	0.3648	Correlation preservation score: 99.29/100 .
Spearman preservation (rank)	—	Spearman preservation score: 98.10/100 .
Mutual information difference (mean)	N/A	N/A.
Detectability — classifier AUC	0.991467	Indistinguishability score: 1.71/100 — classifier easily separates real vs synthetic.
Detectability — average precision (AP)	0.991538	AP-based indistinguishability: 0.85/100 (strong signal).
Propensity gap (mean absolute from 0.5)	0.400392	Propensity gap score: 19.92/100 (larger gap = easier to detect).

Metric	Value	One-line interpretation
Average nearest-neighbor distance (synth→real)	0.171147 (scaled units)	NN distance indicates synthesized records are not exact copies.
NN non-copying heuristic score	100.0	Heuristic indicates low copying risk by distance metric.
Copy risk fraction (very-close threshold)	0.0	No synthetic rows fall into the conservative “very-close” copy bucket.
Copy risk score	100.0	Copy-risk friendly score: 100/100 .
Exact duplicate fraction	0.0	No exact duplicates — excellent.
Re-identification (quasi-identifiers) match fraction	0.0	No unique matches on tested quasi-identifiers — excellent.
Synthetic uniqueness fraction	0.994	99.4% unique rows — high diversity.
Real range coverage (mean across numerics)	0.950813	Range coverage score: 95.08/100 .
Class balance TV / similarity	N/A	No labelled target provided (N/A).
Overall combined quality (0–100)	94.55	Weighted summary of core metrics.

What this means for your team

- **Data science readiness:** Strong suitability for model prototyping, feature engineering, and analytics — preserved means, spreads, and correlations make downstream behavior reliable.
- **Collaboration and demos:** Safe to share internally and for controlled partner demos. Privacy checks (duplicates, copy-risk, re-ID) show no direct matches.
- **External sharing:** If the client requires near-perfect indistinguishability (classifier AUC close to 0.5), a short targeted hardening step is recommended prior to broad external sharing.

Privacy posture (concise)

- **Exact-copy checks:** 0% duplicates found.
- **Copy-risk checks:** 0% of synthetic rows fell below our conservative “very-close” threshold.
- **Re-identification test (quasi-identifiers):** 0% of synthetic rows matched uniquely to a real row on the tested quasi-identifiers.
- **Net result:** On the tests performed, the synthetic sample demonstrates a strong privacy posture for typical sharing scenarios. For highly regulated contexts, we can provide a hardened variant and re-run checks.

Short glossary (plain language)

Distribution match (JS / KS / Wasserstein) — measures whether feature histograms and supports look alike between real and synthetic.

Mean / Std / Median — basic summaries (average, spread, middle), important for business KPIs and aggregations.

Correlation / Rank — whether relationships between features (linear and order-preserving) are preserved.

Detectability (AUC / AP) — how well a simple model can separate real from synthetic records; AUC near 0.5 is ideal.

Copy risk / Re-ID — checks ensuring synthetic rows aren't copies and do not reproduce unique combinations of quasi-identifiers.